

# ENIGMA NUMBER

GitHub  
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+ h e r o h a n k a r t h i k e y a  
20, 8, 5, 18, 15, 8, 1, 18, 11, 1, 18, 20, 8, 9, 11, 5, 25, 1

## # MY ENIGMA NUMBER

$$20 \times 2 + 8 \times 4 + 5 \times 8 + 18 \times 16 + 15 \times 32 + 8 \times 64 + 1 \times 128 + 18 \times 256 + 5 \times 512 + 1 \times 1024$$

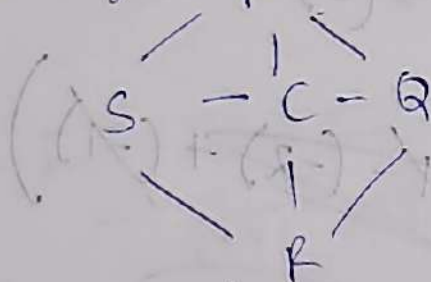
$$+ 20 \times 2048 + 8 \times 4096 + 9 \times 8192 + 11 \times 16384 + 5 \times 32768 + 65536 \times 25 + 131072$$

$$\Rightarrow 20 + 16 + 20 + 144 + 240 + 256 + 64 + 1792 + 2816 + 512 + 181432 + 40960 +$$

$$32768 + 73728 + 180224 + 163840 + 1638400 + 31072$$

$$E \Rightarrow 2, 185, 304$$

LEVEL  
# 1A: The bridges of your city



Initial Roadmap

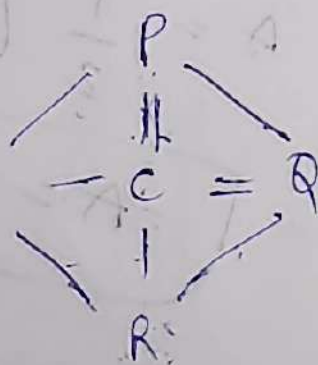
last 4 digits of E: 5304

$$P-C : 5$$

$$Q-C : 3$$

$$R-C : 0$$

$$S-C : 4$$



Revised Road Map

4 outer bridges + 2 double bridges  
+ 2 bridge to center

Q) How many bridges are there in total?

A: 10

Q) Do you think the walk is possible? (Guess)

A: No

$$\Rightarrow 4 + 4 + 2 = 10$$

Attempts:

1)  $P-C \rightarrow C-P \rightarrow P-S \rightarrow S-R \rightarrow R-Q \rightarrow Q-C \rightarrow C-Q \rightarrow Q-P$

2)  $C-S \rightarrow S-P \rightarrow P-Q \rightarrow Q-R \rightarrow R-C \rightarrow C-Q \rightarrow Q-C \rightarrow C-P \rightarrow P-C \rightarrow C-S$   
 $\rightarrow S-R$  ~~Q-R~~

3)  $R-C \rightarrow C-P \rightarrow P-Q \rightarrow Q-C \rightarrow C-S \rightarrow S-P \rightarrow P-C \rightarrow C-R \rightarrow R-S$

Attempt Analysis:

1) There are no paths left for P to continue

2) There are no paths left for R to continue

3) There are no paths left for S to continue

COUNT:

| Islands | Count |
|---------|-------|
| C       | 6     |
| P       | 4     |
| Q       | 4     |
| R       | 3     |
| S       | 3     |

C, P, Q  $\Rightarrow$  Even counts

R, S  $\Rightarrow$  Odd counts

Decide:

POSSIBLE PATH:

$R \rightarrow S \rightarrow P \rightarrow C \rightarrow P \rightarrow Q \rightarrow C \rightarrow Q \rightarrow R \rightarrow C \rightarrow S$  ~~Q-R~~

IMPOSSIBLE PATH:

$R \rightarrow C \rightarrow P \rightarrow Q \rightarrow C \rightarrow S \rightarrow P \rightarrow C \rightarrow R \rightarrow S$

(6) ~~This path is incomplete as R-Q portion is not being covered.~~  
 There is no bridge for S to continue as the paths are already covered.

Question 6: Why do the counts decide everything?

A: For even counts Islands, there will be both arrival and departure  
 whereas for odd count Islands, there will be <sup>only</sup> arrival, ~~stop~~ and an  
 odd path left to complete the bridge.